



Software Engineering Department
Braude College of Engineering

Final Project – Phase A (61998)

**Robust Remote Heart Rate Estimation under
Artificial Lighting Conditions**

26-1-R-25

By:

Tom Biton — tom.biton@e.braude.ac.il
Tomer Roll — tomer.roll@e.braude.ac.il

Advisor:

Dr. Dan Lemberg — leMBERGDAN@braude.ac.il
Mrs. Elena Kramer — elenak@braude.ac.il

Link to GitHub: https://github.com/tomerroll/Tom_Tomer_FinalProject

June 13, 2026

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1 Abstract

Remote photoplethysmography (rPPG) enables contactless heart rate measurement by analyzing subtle color variations in facial skin captured through a camera. While this technique shows promising results under natural daylight, its reliability significantly deteriorates in indoor environments with artificial lighting. Flickering, uneven illumination, and spectrum distortions introduced by artificial light sources often obscure the pulse-related signal components and reduce the accuracy of heart rate estimation.

This project aims to enhance an existing rPPG system by investigating and integrating advanced signal and image processing techniques to suppress lighting-induced noise. We propose the application of chrominance-based projection methods (such as POS), Retinex filtering for illumination normalization, and other frequency domain filters to improve the signal-to-noise ratio under artificial lighting. Through iterative experimentation and comparative evaluation, we aim to adapt the system to function robustly in indoor scenarios using only standard video recordings from smartphone cameras.

2 Introduction

Accurate heart rate monitoring is a fundamental tool in modern healthcare, particularly for elderly patients in nursing homes or remote settings where continuous contact-based monitoring may be impractical or unavailable. In recent years, contactless pulse detection through remote photoplethysmography (rPPG) has gained attention for its simplicity, accessibility, and potential for integration in camera-based systems.

A previous project developed an rPPG system that detects the subject's forehead region in a video and uses frequency analysis of the green channel to estimate pulse rate. This approach demonstrated good accuracy in outdoor environments using natural daylight. However, the system failed to perform reliably under indoor artificial lighting. Signal instability, spectral artifacts, and high-frequency noise introduced by light sources such as fluorescent or LED lamps interfered with the detection of the true pulse signal. This limitation currently restricts the use of rPPG in many real-world environments where artificial lighting is the norm.

Our project continues this work with a research-oriented goal: to explore and implement signal enhancement methods that will enable accurate heart rate detection under artificial light conditions. We focus specifically on image-based filtering techniques that can be applied after the video is captured, including color projection models and illumination compensation algorithms. By building upon the original rPPG framework and addressing its limitations, we aim to make the technology more robust and applicable to realistic, indoor use cases.

3 Literature Review

3.1 Challenges of Artificial Lighting in rPPG

Remote photoplethysmography (rPPG) is a widely researched approach for estimating heart rate without physical contact, relying on the analysis of subtle color changes in the skin captured through video. These variations are caused by periodic changes in blood volume synchronized with the cardiac cycle. While rPPG has shown promising performance under natural lighting, its accuracy and stability significantly degrade under artificial lighting conditions due to flickering, uneven illumination, and spectral distortions.

Artificial lighting introduces several sources of noise that interfere with accurate pulse detection. Firstly, many indoor lights such as LEDs and fluorescent bulbs emit light that fluctuates at frequencies related to the AC power supply (typically 50 or 60 Hz), resulting in subtle periodic changes in the recorded frames. These fluctuations can overlap or mask the frequency band of the pulse signal, making it harder to isolate. Secondly, artificial light often lacks the full spectrum of daylight, leading to color imbalances that reduce the consistency of skin tone tracking across frames. In addition, reflections and shadowing caused by artificial lighting create local intensity variations across the face, which further degrade signal quality. These effects collectively reduce the signal-to-noise ratio (SNR) of the extracted photoplethysmographic signal, making conventional rPPG techniques less reliable indoors.

3.2 Temporal Signal Filtering Techniques

To address these limitations, various filtering and signal processing strategies have been developed and evaluated in the literature. One foundational method for improving signal quality in rPPG is the application of band-pass filters. These filters isolate frequency components corresponding to the human heart rate range, typically between 0.7 and 4 Hz. This approach eliminates low-frequency drifts caused by ambient light fluctuations and high-frequency noise originating from fluorescent or LED flicker. A common implementation uses the Butterworth filter, which provides smooth passband characteristics and is computationally efficient using SciPy’s signal processing tools.

3.3 Chrominance Based Methods: CHROM and POS

Another effective strategy is color projection-based filtering. The CHROM method calculates orthogonal chrominance signals from normalized RGB values and constructs a pulse signal by balancing the standard deviations of these signals. This technique enhances pulsatile color variations while reducing the influence of global illumination.

Similarly, the POS method projects the normalized color signal onto a matrix designed to suppress shared variations across the RGB channels, which are typically caused by lighting changes. Both methods have shown improved performance in dynamic environ-

ments and are implemented using standard matrix operations and temporal normalization procedures.

3.4 Illumination Normalization with Retinex

Image preprocessing approaches also play a critical role in mitigating artificial light interference. Retinex-based algorithms, such as Single Scale Retinex (SSR) and Multi Scale Retinex (MSR), were originally developed for illumination correction in image enhancement. These methods decompose each frame into illumination and reflectance components, suppressing intensity gradients while preserving local contrast. In the context of rPPG, applying Retinex filters to each frame before signal extraction can significantly reduce lighting-induced noise and enhance the consistency of skin tone detection across time.

3.5 Blind Source Separation Using ICA

In more advanced signal domain techniques, Independent Component Analysis (ICA) has been widely used to extract the pulse signal from the RGB channels. ICA operates under the assumption that the observed color signals are linear mixtures of independent sources, including the pulse and various noise components. By maximizing the statistical independence of components, ICA isolates the dominant periodic signal, presumed to correspond to the heart rate. This method is particularly effective when applied after preliminary filtering, and it is typically implemented using signal whitening and eigenvalue decomposition techniques available in libraries such as Scikit-learn or SciPy.

3.6 Frequency Domain Filtering with CDF

An additional method that focuses on suppressing color distortion in the frequency domain is the Color Distortion Filter (CDF). This approach projects each frequency component of the RGB signal into an orthogonal space designed to capture lighting-induced distortions. It then attenuates components with high distortion and retains those aligned with the pulse signature. The method has been shown to improve signal quality in challenging environments, including those with fluorescent or halogen lighting.

3.7 Low Rank Component Reconstruction: LA-SSA

Lastly, Singular Spectrum Analysis combined with Low Rank Matrix Decomposition, referred to as LA-SSA, has been introduced as a comprehensive denoising technique. This method first decomposes the signal into singular components and identifies those with strong periodicity using autocorrelation. The selected components are reconstructed using low-rank approximation to suppress non-periodic noise. LA-SSA has demonstrated

improved robustness in scenarios involving motion and lighting instability and can be implemented using numerical decomposition libraries.

3.8 Summary

Collectively, these filtering and enhancement techniques represent the most prominent directions in addressing the challenges of artificial lighting in rPPG systems. Each method contributes a specific strategy—whether through temporal filtering, chromatic projection, illumination normalization, or component decomposition—and many of them can be implemented without deep learning models, relying solely on classic image and signal processing tools. This makes them particularly well suited for integration into traditional rPPG pipelines such as the one used in this project.

4 System Requirements

4.1 Functional Requirements

- The system shall accept a video file recorded under artificial lighting conditions as input.
- The system shall extract frames from the video and detect the forehead region (ROI) for each frame.
- The system shall support accurate heart rate estimation despite the presence of artificial lighting noise.
- The system shall display the extracted heart rate in beats per minute (BPM).
- The system shall incorporate signal filtering algorithms to remove noise caused by artificial lighting.
- The system shall provide an alert when abnormal heart rate values are detected.
- The system shall extract average RGB color values from the ROI region in each video frame.

4.2 Non-Functional Requirements

- Platform Compatibility: The system shall support video input recorded using standard smartphone cameras, operating correctly with frame rates ranging from 24 to 60 FPS.
- Modifiability: The codebase shall support easy substitution or improvement of filtering components without affecting unrelated modules.

- **Extensibility:** It shall be possible to add new illumination correction methods or pulse detection algorithms in the future with minimal architectural changes.
- **Testability:** Each core functionality shall be verifiable through unit tests.
- **Maintainability:** The system shall include internal documentation and comments explaining the filtering logic and configuration parameters for future developers.
- **Efficiency:** The integration of signal enhancement algorithms (e.g., POS, Retinex) shall not increase total processing time by more than 25% compared to baseline performance.
- **Integrability:** All filtering methods shall be modular and compatible with the existing signal processing pipeline without requiring fundamental restructuring.
- **Usability:** The system interface shall allow non-expert users (e.g., medical staff or researchers) to operate the video upload and view results without prior training.

5 Requirements Gathering Description

In order to thoroughly understand the structure, limitations, and context of the original rPPG project upon which our work is based, we conducted a structured telephone interview with the system’s original developer. This conversation aimed to clarify technical details, identify known challenges, and explore previously attempted solutions, particularly regarding the system’s performance under artificial lighting conditions.

During the interview, the developer provided a detailed overview of the system’s architecture. The original implementation accepts a video file as input, segments it into frames, detects the forehead region (ROI) in each frame using classical computer vision techniques, extracts average color values (RGB) from the ROI, and then analyzes the resulting signal using frequency domain analysis (e.g., Fast Fourier Transform) to estimate the heart rate.

A significant portion of the discussion focused on the observed limitations of the system in indoor environments. The interviewee explained that the system had shown acceptable results when tested under natural daylight but consistently failed to produce accurate or repeatable heart rate estimates when used indoors with artificial lighting. These inaccuracies were initially unpredictable and varied significantly across recordings. However, over time, the team identified artificial light flickering and spectral distortion as primary sources of error that degraded the signal quality.

When asked about mitigation efforts, the developer confirmed that the original team had experimented with basic signal filters to reduce noise, but did not adopt any specialized, literature-based methods for compensating for illumination artifacts. Their focus remained on establishing the core rPPG pipeline, and dealing with artificial lighting was not a primary objective during their project phase.

The insights gained from this interview directly informed the research direction of our extended project. Understanding both the structural framework and the unresolved challenges allowed us to define clear goals for enhancement—particularly the integration of validated signal processing methods aimed at improving performance under artificial lighting.

6 Research and Development Process Description

The project will begin with a thorough analysis of the baseline system upon which our work builds. This will include conceptual understanding through discussions with the original developer, theoretical exploration through the study of academic literature on remote photoplethysmography (rPPG), and practical analysis by reviewing and testing the existing codebase. We will also evaluate the current system from a user perspective to experience its behavior and limitations in real-world usage scenarios.

Following this initial phase, the focus will shift toward identifying and characterizing the system’s limitations under artificial lighting conditions. This step includes gathering insights from the original developer regarding previously attempted solutions, investigating the technical root causes of performance degradation due to artificial illumination, and formally defining the problem based on observations and prior experiments.

Once the problem is clearly defined, we will conduct a comprehensive literature review to identify existing signal and image processing techniques specifically designed to mitigate lighting-induced distortions in rPPG systems. The identified methods will be critically analyzed based on their theoretical foundation, reported effectiveness, computational efficiency, and compatibility with the architecture of the current system.

From this review, we will select a subset of methods with the highest potential to enhance the robustness of heart rate estimation under artificial lighting. Each candidate method will be evaluated for practical feasibility, including implementation complexity, expected runtime overhead, and integration with the existing pipeline.

This stage will culminate in the implementation, testing, and comparison of the selected methods, aiming to validate their effectiveness in improving signal quality and accuracy in indoor environments. Throughout the project, the development process will be iterative and research-driven, balancing theoretical rigor with engineering constraints, and continuously refining the approach based on empirical findings.

7 Expected Challenges

7.1 Algorithm Implementation Complexity

Several of the filtering and signal enhancement techniques identified in the literature—such as POS, Retinex (SSR/MSR), and Color Distortion Filtering (CDF)—involve non-trivial

mathematical operations, frequency domain transformations, or color space manipulations. Implementing these methods correctly while preserving numerical stability and minimizing information loss will require a deep understanding of both the algorithms and the structure of the input video signal.

7.2 Integration with Existing Codebase

The current rPPG system follows a specific processing pipeline, from frame extraction through ROI detection to signal analysis. Integrating new modules into this workflow—especially image-based preprocessing methods—may introduce compatibility issues. Aligning color normalization stages, ensuring correct signal flow, and maintaining modularity in the system will pose technical challenges.

7.3 Ensuring Real Time or Near Real Time Performance

While the added filtering steps are designed to improve robustness under artificial lighting, they also add computational overhead. Ensuring that the system can still operate with reasonable latency, particularly when using smartphone-grade hardware or standard computing environments, is a significant concern. Optimization strategies or trade-offs between accuracy and speed may need to be considered.

7.4 Testing and Validation under Diverse Lighting Conditions

Evaluating the effectiveness of the improvements requires capturing and testing video data under a wide range of artificial lighting scenarios (e.g., LED, fluorescent, halogen). This involves careful experimental design, consistent ground truth measurements for comparison, and possibly developing custom testing tools or environments to validate the improvements.

7.5 Signal Ambiguity and Interference

In indoor settings, the pulse signal can often be masked by dominant low-frequency or high-frequency noise components from lighting flicker, reflections, or skin tone variations. Disentangling true physiological signals from environmental noise—especially in the absence of ground truth—is an inherent difficulty that may impact accuracy and evaluation.

8 Tools and Technologies

In the development and research of this project, we intend to utilize the following set of tools and technologies:

8.1 Development Environment

- PyCharm: As the primary integrated development environment (IDE) for writing, testing, and debugging the Python-based codebase.
- Python 3.x: The main programming language used for implementing signal and image processing algorithms.

8.2 Version Control and Collaboration

- Git: For tracking changes and maintaining version history of the codebase.
- GitHub: For centralized repository hosting, collaboration, and backup of project files.

8.3 Algorithmic Libraries and Scientific Computation

To implement and evaluate the signal and image processing methods (especially POS, CHROM, and Retinex), we will use the following libraries:

- OpenCV: For image processing tasks such as color space conversion, filtering, ROI extraction, and Retinex-based enhancement.
- NumPy & SciPy: For numerical computations, array manipulation, filtering (e.g., Butterworth), and frequency analysis (FFT).
- Matplotlib / Seaborn: For visualizing signals and heart rate estimation results.
- scikit-image: For additional image enhancement and preprocessing routines, including optional implementations of Retinex filters.

8.4 Testing and Experimentation

- Smartphone Camera: As the standard acquisition tool to record test videos under both natural and artificial lighting scenarios.

9 Description of the Solution – Software Architecture and Operation

9.1 Existing System Architecture

The current rPPG system is designed to estimate heart rate from a recorded facial video using a conventional processing pipeline. The main stages of the existing architecture are as follows:

- **Video Input:** The user uploads or captures a video via the application interface.
- **ROI Detection:** The system detects the face and isolates the forehead region (Region of Interest – ROI) from each frame.
- **Color Signal Extraction:** For each ROI, the system calculates the median value of the green channel (G) over time, based on the assumption that the green channel carries the strongest pulsatile signal.
- **Signal Buffering:** A rolling buffer stores a predefined number of green channel samples (e.g., 600 samples).
- **Frequency Analysis:** The system applies Fast Fourier Transform (FFT) to the G channel signal.
- **Heart Rate Estimation:** The frequency domain is scanned for the highest peak within a plausible range (e.g., 0.7–4 Hz), and the corresponding frequency is translated into beats per minute (BPM).
- **Output Display:** The BPM value is visualized in the user interface.

This implementation works well under stable natural lighting, but its accuracy deteriorates significantly under artificial lighting due to illumination fluctuations and spectral noise.

9.2 Proposed Architecture with Enhanced Illumination Robustness

To address the limitations under artificial lighting, the enhanced system will integrate advanced image processing techniques into the existing pipeline. The core idea is to preprocess the ROI signal to suppress illumination-related noise while preserving the true pulsatile component. Two experimental configurations will be tested:

9.2.1 Retinex + POS

Retinex Filtering is applied first to each ROI frame. This technique normalizes illumination by decomposing each image into reflectance (intrinsic object color) and illumination (external light). We apply either Single Scale Retinex (SSR) or Multi Scale Retinex (MSR) to reduce uneven lighting and enhance local contrast in the skin region. After Retinex, we apply the POS algorithm (Plane Orthogonal to Skin tone), which performs a projection of normalized RGB signals to a space that suppresses color changes shared across all channels (e.g., due to lighting). POS isolates the pulse-induced color variation by computing weighted projections based on chrominance dynamics.

Integration point:

- Retinex is inserted before calculating average color values in each ROI.
- POS replaces the direct G-channel extraction and is applied to the RGB time series.

Advantages:

- Retinex enhances signal uniformity under varying lighting.
- POS further suppresses residual global illumination changes.
- Combined, these methods enhance signal-to-noise ratio without changing system output structure.

9.2.2 Retinex + CHROM

In this configuration, Retinex preprocessing remains the same and is used to suppress spatial illumination inconsistencies. Then, instead of POS, we apply the CHROM algorithm, which was proposed by De Haan and Jeanne. CHROM constructs a specific linear combination of R, G, and B channels to remove global lighting variations based on chrominance vectors. Unlike POS, CHROM includes dynamic weighting (via standard deviation ratio of components) to adaptively cancel lighting artifacts.

Integration point:

- Retinex filtering is applied as a pre-step to ROI frames.
- The CHROM method is used to calculate the pulse signal from normalized RGB sequences, replacing G-channel based signal formation.

Advantages:

- CHROM offers more sophisticated compensation for dynamic lighting changes.

- When combined with Retinex, this method is expected to yield greater robustness in fluctuating indoor environments such as under fluorescent or LED lighting.

Both configurations are designed to be modular and fit into the original data flow. Their insertion occurs before frequency analysis, maintaining compatibility with the rest of the pipeline (FFT, BPM calculation, and display).

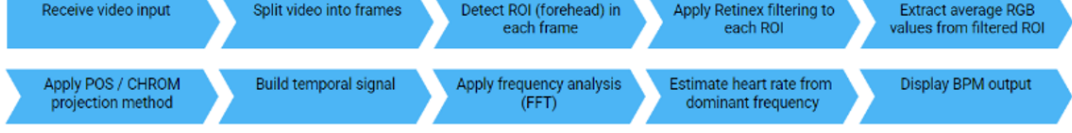


Figure 1: flow process

Flowchart Description:

Receive video input → Split video into frames → Detect ROI (forehead) in each frame → Apply Retinex filtering to each ROI → Extract average RGB values from filtered ROI → Apply POS / CHROM projection method → Build temporal signal → Apply frequency analysis (FFT) → Estimate heart rate from dominant frequency → Display BPM output.

10 Success Metrics

The primary objective of this project is to enhance the accuracy and robustness of heart rate estimation using remote photoplethysmography (rPPG) under challenging artificial lighting conditions. To objectively assess the extent to which this goal has been achieved, the following quantitative metrics will be used:

10.1 Mean Absolute Error (MAE) of Heart Rate Estimation

The heart rate values estimated by the system will be compared to ground truth measurements obtained from a contact-based reference device (e.g., pulse oximeter). The Mean Absolute Error (MAE), expressed in beats per minute (BPM), will serve as the principal accuracy metric. According to widely accepted thresholds in rPPG literature, $MAE \leq 5$ BPM is considered acceptable for indoor environments, while values in the range of 2–4 BPM are regarded as optimal for non-contact applications.

10.2 Signal to Noise Ratio (SNR)

To quantify signal quality, the Signal to Noise Ratio (SNR) of the extracted rPPG signal will be computed in decibels (dB) after applying frequency domain analysis. Prior studies have established that an SNR above 3–5 dB is required for meaningful pulse detection, while values above 6 dB are indicative of high fidelity measurements. A successful implementation will demonstrate an improvement of at least 2 dB in SNR compared to the raw green channel baseline.

10.3 Robustness to Lighting Variability

The system will be evaluated under multiple artificial light sources, including fluorescent, LED, and incandescent lighting, as well as under varying intensities and angles. To be considered robust, the system’s MAE should vary by no more than ± 2 BPM across these conditions. This criterion is derived from literature reports that indicate such variance remains within acceptable physiological and clinical tolerances.

10.4 Performance Over Baseline Method

The improved system will be benchmarked against the original implementation that estimates heart rate using the green channel without filtering. A performance improvement of 30–50% in either MAE or SNR (or both) will be the minimum requirement to consider the integrated signal enhancement strategies effective.

10.5 Real Time Feasibility

The system must retain computational efficiency to support near real-time use. Specifically, the average processing time per frame (including all filtering and analysis steps) must not exceed a 25% increase over the baseline implementation. Furthermore, the system must be able to process at least 15 frames per second (FPS) to maintain responsiveness during typical video processing tasks.

The numeric thresholds outlined above are grounded in empirical evidence drawn from prior peer-reviewed publications in the field of remote PPG, including works by Wang et al. (2017), de Haan and Jeanne (2013), and subsequent studies that evaluated signal quality under variable lighting. These values reflect commonly accepted benchmarks for clinical relevance, signal fidelity, and computational practicality in non-contact vital sign monitoring systems.

11 Testing Methodology

To ensure that the extended rPPG system functions correctly and meets the defined objectives, a structured testing process will be implemented. Since this project builds upon a previously developed system, the testing efforts will primarily focus on the newly introduced components and their integration into the existing pipeline. The testing process will include the following phases:

11.1 Unit Testing of New Components

Unit testing will be applied exclusively to the modules developed as part of this project extension. These include:

- The implementation of illumination normalization algorithms (e.g., Single-Scale Retinex, Multi-Scale Retinex).
- The chrominance-based signal extraction methods (POS and CHROM).
- The updated signal filtering and preprocessing logic introduced to mitigate artificial lighting noise.

Each function will be tested in isolation using controlled inputs and validated outputs to ensure algorithmic correctness and numerical stability. Tests will be written using Python’s unittest or pytest frameworks.

11.2 Integration Testing

Once individual modules are verified, integration testing will confirm the compatibility of the new components with the existing processing pipeline. This phase will evaluate:

- Whether filtered ROI frames are correctly passed through to the chrominance projection algorithms.
- The smooth incorporation of illumination-normalized signals into the heart rate estimation logic.
- The maintenance of correct timing, signal alignment, and consistency across pipeline stages.

Integration tests will simulate realistic end-to-end workflows but will focus on scenarios that exercise the newly added logic paths.

11.3 System-Level Testing

Full-system testing will validate that the complete application—from video input to heart rate display—functions reliably. These tests will incorporate both existing modules and new filtering components to evaluate the overall correctness, output stability, and user-facing behavior. This includes:

- Verifying that heart rate estimates are presented accurately in BPM.
- Ensuring abnormal heart rate alerts are triggered correctly.
- Confirming correct operation with typical input videos from smartphones.

11.4 Performance Evaluation

The performance impact of the new algorithms will be quantified in terms of:

- Processing time per frame (target: $\leq 25\%$ increase over baseline).
- Frame throughput (target: ≥ 15 FPS on standard hardware).
- Memory usage and CPU efficiency using profiling tools.

This ensures that the system remains responsive and practical even after enhancement.

11.5 Accuracy and Robustness Testing

To evaluate signal quality and practical effectiveness, the system will be tested using videos recorded under various artificial lighting conditions. Metrics will include:

- Mean Absolute Error (MAE) in BPM compared to pulse oximeter ground truth.
- Signal-to-Noise Ratio (SNR) of the extracted waveform.
- Stability of heart rate estimates across different lighting environments (e.g., LED, fluorescent, incandescent).

Controlled experiments will be used to assess how reliably the enhancements address lighting-induced noise and improve the underlying rPPG signal quality.

12 AI Tools Utilized and Representative Prompts

In the research and analysis phase of this project, we leveraged advanced AI tools to support both theoretical investigation and structured extraction of information from documentation. The use of each tool was aligned with its specific capabilities and strengths, as follows:

12.1 ChatGPT (OpenAI)

Purpose of Use: ChatGPT was primarily used to build a strong conceptual understanding of the problem space, especially concerning the challenges of remote photoplethysmography (rPPG) under artificial lighting. It played a central role in:

- Discovering and identifying relevant peer-reviewed literature addressing noise filtering and illumination correction in rPPG.

- Understanding the theoretical foundations and operational mechanisms of signal and image processing techniques discussed in the literature.
- Analyzing the limitations of the original rPPG project in handling artificial light fluctuations, and identifying what algorithmic components were missing or under-developed.
- Exploring the feasibility of combining different approaches (e.g., Retinex + POS/CHROM) to enhance performance.

Example Prompts:

- “What are effective frequency domain filtering techniques for rPPG under indoor lighting?”
- “Explain how CHROM and POS algorithms reduce illumination artifacts in contactless heart rate estimation.”
- “Based on common noise sources in LED-lit environments, how can I preprocess video frames for better pulse extraction?”

12.2 Gemini (Google)

Purpose of Use: Gemini was used to complement ChatGPT’s findings by cross-referencing results and validating concepts. It helped in:

- Identifying and comparing relevant academic papers, including those not initially retrieved by other tools.
- Extracting insights about how different algorithms performed under varying light sources.
- Guiding decision-making regarding which signal enhancement strategies are most practical to incorporate into the existing pipeline.

Example Prompts:

- “Which non-deep learning rPPG methods work well under fluorescent lighting?”
- “Compare Color Distortion Filtering (CDF) and ICA for denoising rPPG signals.”
- “What do recent studies suggest as minimum SNR thresholds for reliable pulse estimation indoors?”

12.3 NotebookLM (Google)

Purpose of Use: NotebookLM was primarily used to extract structured, document-specific insights from the base project report and academic papers. It enabled:

- Retrieval of exact methods used in prior work, including preprocessing pipelines, ROI strategies, and pulse estimation procedures.
- Extraction of performance metrics (e.g., MAE, SNR) from previously conducted experiments.
- Summarizing comparative advantages and limitations of each method mentioned in uploaded literature or project documentation.

Example Prompts:

- “From the attached project report, what is the exact flow of heart rate estimation from video input to output?”
- “What signal filtering algorithms were applied in the reference capstone system?”
- “According to the uploaded article, what are the accuracy benchmarks for rPPG systems under LED lighting?”

This multi-tool approach allowed us to combine exploratory understanding (via ChatGPT and Gemini) with precise fact extraction (via NotebookLM), providing a well-rounded foundation for system design, algorithm selection, and strategic integration of enhancements tailored to artificial lighting scenarios.

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